Chapter 6

Joint distributions, marginal distributions, and conditional distributions

We have seen in Section 2.4 that if random variables X_1 and X_2 are *independent*, then we can use *independence* of X_1 and X_2 to compute

$$\mathbb{P}(X_1 \in A_1,\ X_2 \in A_2) = \mathbb{P}(X_1 \in A_1)\mathbb{P}(X_2 \in A_2)$$

If X_1 and X_2 are not independent, how can we compute $\mathbb{P}(X_1 \in A_1,\ X_2 \in A_2)$?
In many practical situations, random variables are not independent. To compute joint probabilities in such cases, we need to understand their joint distribution.

Example 6.1. Suppose we select a student at random and let

X = student's height in cm

Y = student's weight in kilograms

Intuitively, it is clear that X and Y are not independent (because if you know the value of X, the student's height, then you are better able to predict the student's weight Y than you would be without knowing the value of X).

In order to compute joint probabilities such as

$$\mathbb{P}(68 < X \leqslant 72,\ 65 < Y \leqslant 75)$$

for the variables in the example above, we need to know the joint distribution of X and Y.

6.1 Joint probability mass function for discrete random variables

For a discrete random variable X, the simplest way to express the distribution is through the probability mass function $p_X(x) = \mathbb{P}(X = x)$ for $x \in \mathrm{Im}X$. To deal with two (or more) discrete random variables at once, we can use the joint probability mass function.

Definition. Suppose X and Y are discrete random variables with $\mathrm{Im}X = \{x_1, x_2, \dots\}$ and $\mathrm{Im}Y = \{y_1, y_2, \dots\}$. Then the **joint probability mass function** $p_{X,Y}(x, y)$ is defined to be

$$\boxed{p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y) \qquad \forall x \in \mathrm{Im}X,\ y \in \mathrm{Im}Y}$$

Properties of joint probability mass functions

Suppose that $p_{X,Y}$ is the joint probability mass function of (discrete) random variables X, Y. Then:

1. $p_{X,Y}(x, y) \geqslant 0 \ \forall x, y$
2. $\sum_x \sum_y p_{X,Y}(x, y) = 1$

Conversely, any function satisfying the above properties is a valid joint probability mass function for some pair of discrete random variables.

Example 6.2. Suppose a fair coin is flipped 4 times. Define

X = number of Heads in first 3 flips

Y = number of Heads in last 2 flips

It seems intuitively clear that X and Y will not be independent here. To check this, we first find the joint probability mass function.

We have $\text{Im}X = \{0, 1, 2, 3\}$ and $\text{Im}Y = \{0, 1, 2\}$.

The sample space Ω consists of the 16 possible outcomes of flipping a coin 4 times. The coin is fair, and it is reasonable to assume that flips are independent of one another, so the 16 possible outcomes are all equally likely. So for each (x, y) with $x \in \text{Im}X$, $y \in \text{Im}Y$ we can find the probability $\mathbb{P}(X = x, Y = y)$ by counting the number of outcomes that result in x Heads in the first 3 flips and y Heads in the last 2 flips.

For instance, $p_{X,Y}(1, 1) = \mathbb{P}(X = 1, Y = 1) = \mathbb{P}(\{HTTH,\ THTH,\ TTHT\})$

We can present the joint probability mass function in a table:

$p_{X,Y}(x, y)$	$x = 0$	$x = 1$	$x = 2$	$x = 3$	
$y = 0$	$1/16$	$2/16$	$1/16$	0	$p_Y(0) = 1/4$
$y = 1$	$1/16$	$3/16$	$3/16$	$1/16$	$p_Y(1) = 1/2$
$y = 2$	0	$1/16$	$2/16$	$1/16$	$p_Y(2) = 1/4$
	$p_X(0) = 1/8$	$p_X(1) = 3/8$	$p_X(2) = 3/8$	$p_X(3) = 1/8$	

Exercise: By enumerating possible outcomes (HHHH, HHHT, etc.), check that the values in the above joint probability mass function table are correct.

Exercies: Considering X and Y separately, check that $X \sim \text{Bin}(3, 1/2)$ and $Y \sim \text{Bin}(2, 1/2)$. Note that this tells us nothing about the relationship *between* X and Y.

The entries in the table above give the values of the joint probability mass function $p_{X,Y}$. By summing up the rows and columns, we can use this table to compute the *marginal* probability mass functions p_X and p_Y for the random variables X and Y, respectively. The two marginal probability mass functions p_X and p_Y are presented in the margins of the table above.

Marginal probability mass functions

Given the joint probability mass function $p_{X,Y}$ for discrete random variables X and Y, we can find the (*marginal*) probability mass functions p_X and p_Y for X and Y as follows:

Suppose $x \in \text{Im}X$, then

$$p_X(x) \underset{\text{def.}}{=} \mathbb{P}(X = x) = \mathbb{P}\Big(\{X = x\} \cap \big(\cup_{y \in \text{Im}Y} \underbrace{\{Y = y\}}_{\substack{\text{these events} \\ \text{form a partition}}} \big)\Big)$$

$$\begin{aligned} &= \mathbb{P}\big(\cup_{y \in \text{Im}Y} \{X = x,\ Y = y\}\big) && \text{distributive property from set theory} \\ &= \sum_{y \in \text{Im}Y} \mathbb{P}(X = x,\ Y = y) && \text{property (III) of } \mathbb{P} \\ p_X(x) &= \sum_{y \in \text{Im}Y} p_{X,Y}(x, y) \end{aligned}$$

Similarly, for $y \in \text{Im}Y$,

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_{x \in \text{Im}X} p_{X,Y}(x, y).$$

Independent discrete random variables

It follows from the definition of independent random variables in Section 2.4 that discrete random variables X and Y with joint probability mass function $p_{X,Y}$ and marginal probability mass functions p_X and p_Y are *independent* if and only if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \qquad \forall x \in \text{Im}X,\ y \in \text{Im}Y$$

Example 6.2 continued. In Example 6.2 we have $p_{X,Y}(0, 2) = 0$, but the corresponding marginal probabilities are $p_X(0) = 1/8$ and $p_Y(2) = 1/4$, so that $p_{X,Y}(0, 2) \neq p_X(0)p_Y(2)$. The random variables X, Y are <u>not</u> independent.

Question: Suppose we know the joint probability mass function $p_{X,Y}$ for variables X and Y, how can we determine whether X and Y are independent?

Answer: To show that X and Y <u>are</u> independent, we need to check that $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for *every* pair of values $x \in \text{Im}X,\ y \in \text{Im}Y$.
To show that X and Y are <u>not</u> independent, it is sufficient to find *any* pair of values $x \in \text{Im}X,\ y \in \text{Im}Y$ with $p_{X,Y}(x, y) \neq p_X(x)p_Y(y)$.

Remarks

1. If we <u>know</u> that discrete random variables X and Y are independent, then we can find their joint probability mass function from their marginal probability mass functions using the relationship $p_{X,Y}(x, y) = p_X(x)p_Y(y)$. Can <u>only</u> do this if we know that X, Y are independent.
2. When the variables X and Y depend on *different experiments*, which are *physically independent*, then it is reasonable to assume that the variables X and Y are (statistically) independent and we can compute $p_{X,Y}$, the joint probability mass function for X and Y, by multiplying their marginal probability mass functions.
3. In general, knowing the marginal probability mass functions $p_X(x)$ and $p_Y(y)$ is <u>not sufficient</u> to be able to find the joint probability mass function $p_{X,Y}(x, y)$.

Example 6.3. Suppose X is a random variable with ImX=$\{1,2\}$ and with probability mass function given by $p_X(1) = p_X(2) = 1/2$ and suppose Y is a random variable with ImY=$\{1,2,3\}$ and with probability mass function given by $p_Y(1) = p_Y(2) = p_Y(3) = 1/3$.

This information gives us the row sums and the column sums for the table that describes the joint probability mass function $p_{X,Y}(x,y)$, but this information doesn't tell us *how* to fill in the table. There are many ways to fill in the table so that the row sums and column sums agree with the values of p_X and p_Y. Here are two possibilities:

	$y=1$	$y=2$	$y=3$	
$x=1$	$1/6$	$1/12$	$1/4$	$1/2$
$x=2$	$1/6$	$1/4$	$1/12$	$1/2$
	$1/3$	$1/3$	$1/3$	

	$y=1$	$y=2$	$y=3$	
$x=1$	$1/12$	$1/6$	$1/4$	$1/2$
$x=2$	$1/4$	$1/6$	$1/12$	$1/2$
	$1/3$	$1/3$	$1/3$	

Both of these tables of values for $p_{X,Y}$ are *consistent* with the given information, and this means that we can't work out $p_{X,Y}(x,y)$ just from knowledge of p_X and p_Y.

Functions of discrete random variables X and Y

Suppose X and Y are discrete random variables with joint probability mass function $p_{X,Y}(x,y)$ and suppose $g : \mathbb{R}^2 \to \mathbb{R}$ is a function. Let $Z = g(X,Y)$, then Z is *also* a discrete random variable.

To determine the probability mass function $p_Z(z)$ for Z, we first need to determine ImZ. Then we can use the joint probability mass function $p_{X,Y}(x,y)$ for X and Y to determine

$$p_Z(z) = \mathbb{P}(Z=z) = \mathbb{P}(g(X,Y)=z) = \sum_{(x,y):g(x,y)=z} p_{X,Y}(x,y) \quad \forall z \in \text{Im}Z.$$

Example 6.2 continued. Suppose $Z = XY$. By considering the *possible* values that X and Y can jointly take, we obtain

$$\text{Im}Z = \{0,1,2,3,4,6\}.$$

Looking at the table for the joint probability mass function $p_{X,Y}$ in example 6.2, we find that the probability mass function $p_Z(z)$ is given by

$$p_Z(6) = \mathbb{P}(Z=6) = \mathbb{P}(XY=6) = \mathbb{P}((X,Y)=(3,2)) = \frac{1}{16}$$
$$p_Z(4) = \mathbb{P}(Z=4) = \mathbb{P}(XY=4) = \mathbb{P}((X,Y)=(2,2)) = \frac{2}{16}$$
$$p_Z(3) = \mathbb{P}(Z=3) = \mathbb{P}(XY=3) = \mathbb{P}((X,Y)=(3,1)) = \frac{1}{16}$$
$$p_Z(2) = \mathbb{P}(Z=2) = \mathbb{P}(XY=2) = \mathbb{P}((X,Y)\in\{(1,2),(2,1)\}) = \frac{1}{16}+\frac{3}{16} = \frac{4}{16}$$
$$p_Z(1) = \mathbb{P}(Z=1) = \mathbb{P}(XY=1) = \mathbb{P}((X,Y)=(1,1)) = \frac{3}{16}$$
$$p_Z(0) = \mathbb{P}(Z=0) = \mathbb{P}(XY=0) = \mathbb{P}((X,Y)\in\{(0,0),(0,1),(0,2),(0,3),(1,0),(2,0)\})$$
$$= \frac{1}{16}+\frac{2}{16}+\frac{1}{16}+0+\frac{1}{16}+0 = \frac{5}{16}$$

(Note that $p_Z(5) = 0$, since $5 \notin \text{Im}Z$.)

Given the probability mass function p_Z for Z, we can compute the expected value $\mathbb{E}(Z)$:

$$\begin{aligned}\mathbb{E}(Z) &= 0 \times p_Z(0) + 1 \times p_Z(1) + 2 \times p_Z(2) + 3 \times p_Z(3) + 4 \times p_Z(4) = 6 \times p_Z(6) \\ &= 0 + \frac{3}{16} + 2 \times \frac{4}{16} + 3 \times \frac{1}{16} + 4 \times \frac{2}{16} + 6 \times \frac{1}{16} \\ &= \frac{28}{16} = \frac{7}{4} = 1.75\end{aligned}$$

Exercise: Compute $\mathbb{V}\text{ar}(Z)$

Also, from the marginal probability mass functions for X and Y we can compute $\mathbb{E}(X) = 3/2$ and $\mathbb{E}(Y) = 1$ (*exercise*: check). Notice that $\mathbb{E}(XY) \neq \mathbb{E}(X)\mathbb{E}(Y)$ in this example.

Expected value of $g(X, Y)$ for discrete X and Y
Just as in the case of a function of a single variable, there is a shortcut for computing $\mathbb{E}\big(g(X, Y)\big)$ which does not require us to first determine the probability mass function for $g(X, Y)$. Specifically:

$$\boxed{\mathbb{E}\left(g(X,Y)\right) = \sum_{\substack{x \in \text{Im}X \\ y \in \text{Im}Y}} g(x,y)\; p_{X,Y}(x,y)}$$

Exercise: Use the above formula to compute $\mathbb{E}(XY)$ in Example 6.2 (you should get the same value as obtained previously).

6.1.1 Conditional probability mass functions

Suppose X and Y are discrete random variables with joint probability mass function $p_{X,Y}$ and suppose y is some specified value such that $p_Y(y) = \mathbb{P}(Y = y) > 0$. Then the **conditional probability mass function** for X *given* $Y = y$ is defined by

$$\boxed{p_{X|Y}(x|y) = \mathbb{P}(X = x | Y = y)}$$

It follows from the definition of conditional probability for *events* that

$$\boxed{p_{X|Y}(x|y) = \mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \frac{p_{X,Y}(x,y)}{p_Y(y)}}$$

Remarks

1. If X and Y are *independent* random variables (and $p_Y(y) > 0$), then

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \overset{\text{by indep of } X \text{ and } Y}{=} \frac{p_X(x)p_Y(y)}{p_Y(y)} = p_X(x)$$

   i.e. whenever random variables are *independent*, the conditional distributions and the unconditional (marginal) distributions for the random variables are the *same*.

2. Sometimes the notation $p_{X|Y}(\cdot \mid y)$ is used, to emphasise that we condition upon one specific value y of Y, and then consider the set of conditional probabilities $p_{X|Y}(x \mid y)$ for all values $x \in \text{Im}X$.

3. Conditional probability mass functions satisfy all the same properties as unconditional probability mass functions.

$$\text{E.g.} \qquad \sum_{x \in \mathrm{Im}X} p_{X|Y}(x|y) = 1$$

4. We can use the conditional probability mass function for X given $\{Y = y\}$ to compute the **conditional expectation of X given $\{Y = y\}$**:

$$\boxed{\mathbb{E}(X \mid Y = y) = \sum_{x \in \mathrm{Im}X} x \; p_{X|Y}(x|y) = \sum_{x \in \mathrm{Im}X} x \; \mathbb{P}(X = x \mid Y = y)}$$

We also have:

$$\mathbb{E}(g(X) \mid Y = y) = \sum_{x \in \mathrm{Im}X} g(x) \; p_{X|Y}(x|y)$$

Example 6.2 continued. Using the definition given above, we can determine the **conditional probability mass function** $p_{X|Y}(x|0)$ of X given $\{Y = 0\}$:

$$p_{X|Y}(0|0) = \frac{p_{X,Y}(0,0)}{p_Y(0)} = \frac{1/16}{1/4} = \frac{1}{4}$$

$$p_{X|Y}(1|0) = \frac{p_{X,Y}(1,0)}{p_Y(0)} = \frac{2/16}{1/4} = \frac{1}{2}$$

$$p_{X|Y}(2|0) = \frac{p_{X,Y}(2,0)}{p_Y(0)} = \frac{1/16}{1/4} = \frac{1}{4}$$

$$p_{X|Y}(3|0) = \frac{p_{X,Y}(3,0)}{p_Y(0)} = \frac{0}{1/4} = 0$$

Note:: The conditional probabilities sum to 1

Using the conditional probability mass function $p_{X|Y}(x|0)$, we can compute the *conditional* expected value of X *given* that $Y = 0$:

$$\mathbb{E}(X \mid Y = 0) = 0 \cdot p_{X|Y}(0|0) + 1 \cdot p_{X|Y}(1|0) + 2 \cdot p_{X|Y}(2|0) + 3 \cdot p_{X|Y}(3|0) =$$
$$= 0 + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot 0 = 1$$

For comparison, the *unconditional* expected value of X is $\mathbb{E}(X) = 3/2$.

Question: Why does it make *sense* that $\mathbb{E}(X) > \mathbb{E}(X \mid Y = 0)$ in this example?

Remark: The conditional expected value $\mathbb{E}(X \mid Y = y)$ is a function that depends on the value of y, so $\mathbb{E}(X \mid Y)$ is a *random variable* that depends on the random variable Y. We can compute

the expected value of the random variable $\mathbb{E}(X \mid Y)$ as follows:

$$\begin{aligned}
\mathbb{E}(\mathbb{E}(X \mid Y)) &= \sum_{y \in \mathrm{Im}(Y)} \mathbb{E}(X \mid Y = y) p_Y(y) \\
&= \sum_{y \in \mathrm{Im}(Y)} \left(\sum_{x \in \mathrm{Im}(X)} x \; p_{X|Y}(x|y) \right) p_Y(y) \\
&= \sum_{y \in \mathrm{Im}(Y)} \sum_{x \in \mathrm{Im}(X)} x \; p_{X,Y}(x,y) \\
&= \sum_{x \in \mathrm{Im}(X)} x \sum_{y \in \mathrm{Im}(Y)} p_{X,Y}(x,y) \\
&= \sum_{x \in \mathrm{Im}(X)} x \; p_X(x) \\
&= \mathbb{E}(X)
\end{aligned}$$

Summary: The calculations above show that $\mathbb{E}(\mathbb{E}(X \mid Y)) = \mathbb{E}(X)$. This result is used frequently in the study of stochastic processes.

6.2 Joint distribution function

Only discrete random variables have (joint) probability mass functions. Just as every random variable X has a cumulative distribution function F_X, *every* pair of random variables X and Y has a joint distribution function $F_{X,Y}(x, y)$, defined as follows.

Definition
Suppose X and Y are random variables. Then the joint distribution function $F_{X,Y} : \mathbb{R}^2 \to [0, 1]$ for X and Y is defined by

$$\boxed{F_{X,Y}(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y) \qquad \forall (x, y) \in \mathbb{R}^2}$$

In general, for random variables $X_1, X_2, \ldots, X_n$, we define the joint distribution function $F_{X_1,X_2,\ldots,X_n} : \mathbb{R}^n \to [0, 1]$ for $X_1, X_2, \ldots, X_n$ by

$$F_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n) = \mathbb{P}(X_1 \leqslant x_1, X_2 \leqslant x_2, \ldots, X_n \leqslant x_n) \qquad \forall (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$$

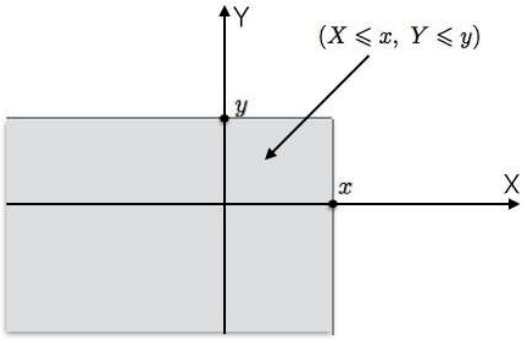
Remarks

1. The definition of joint distribution function $F_{X,Y}(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y)$ is a natural generalisation of the definition of the cumulative distribution function for a single random variable.
2. $F_{X,Y}(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y)$ computes the probability that the ordered pair of variables (X, Y) takes a value in the shaded (infinite) rectangular region:

3. If $x_1 \leqslant x_2$ and $y_1 \leqslant y_2$, then $\{X \leqslant x_1, Y \leqslant y_1\} \subseteq \{X \leqslant x_2, Y \leqslant y_2\}$, so

$$F_{X,Y}(x_1, y_1) \leqslant F_{X,Y}(x_2, y_2)$$

by the monotonicity of $\mathbb{P}$.



4. **Independence of random variables**
   It follows from the definition of independent random variables in Section 2.4 that the random variables $X_1, X_2, ..., X_n$ are independent if and only if for all $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$

$$F_{X_1,X_2,...,X_n}(x_1, x_2, \ldots, x_n) = F_{X_1}(x_1)F_{X_2}(x_2)\cdots F_{X_n}(x_n)$$

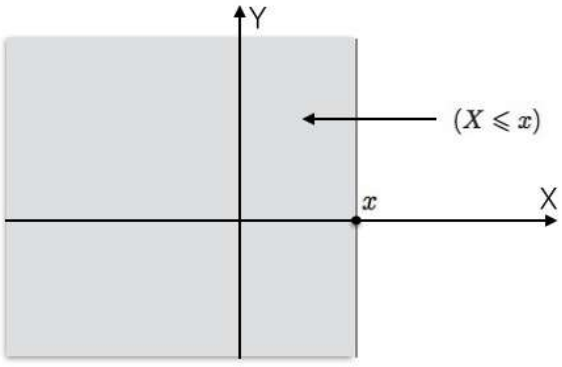
   where $F_{X_1,X_2,...,X_n}$ is the joint distribution function of $X_1, X_2, ..., X_n$ and F_{X_i} is the distribution function of X_i for $i = 1, 2, ..., n$.

Marginal distribution functions

Suppose X and Y have a joint df $F(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y)$ for $(x, y) \in \mathbb{R}^2$. How can we use $F(x, y)$ to find the (marginal) distribution functions of X and Y individually?

Answer: We have

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leqslant x) \\ &= \lim_{y\uparrow\infty} \mathbb{P}(X \leqslant x, Y \leqslant y) && \text{(continuity of } \mathbb{P}) \\ &= \lim_{y\uparrow\infty} F(x, y) && (x \text{ fixed}, y \to \infty) \end{aligned}$$



Likewise, $F_Y(y) = \lim_{x\uparrow\infty} F(x, y)$.

Example 6.4. Suppose X and Y have joint df given by

$$F(x, y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-x-y} & \text{if } x \geqslant 0 \text{ and } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

To find F_X, the (marginal) distribution function of X, we note:

- if $x < 0$, then $F_X(x) = \lim_{y\to\infty} F(x,y) = 0$
- if $x \geqslant 0$, then

$$F_X(x) = \lim_{y\to\infty} F(x,y) = \lim_{y\to\infty} \left(1 - e^{-x} - e^{-y} + e^{-x-y}\right) = 1 - e^{-x}$$

So

$$F_X(x) = \begin{cases} 1 - e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

This is the distribution function of the Exp(1) distribution, so we conclude that $X \sim \text{Exp}(1)$. Similarly, you can show that $Y \sim \text{Exp}(1)$.

In this example, we can see that $F_{X,Y}(x,y) = F_X(x)F_Y(y)$ for every $(x,y) \in \mathbb{R}^2$, so X and Y are **independent** random variables here.